Partial Derivatives and the Gradient

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: First and second partial derivatives

For $f(x, y) = x^3y^2 - 4xy + y$, find:

a f_x

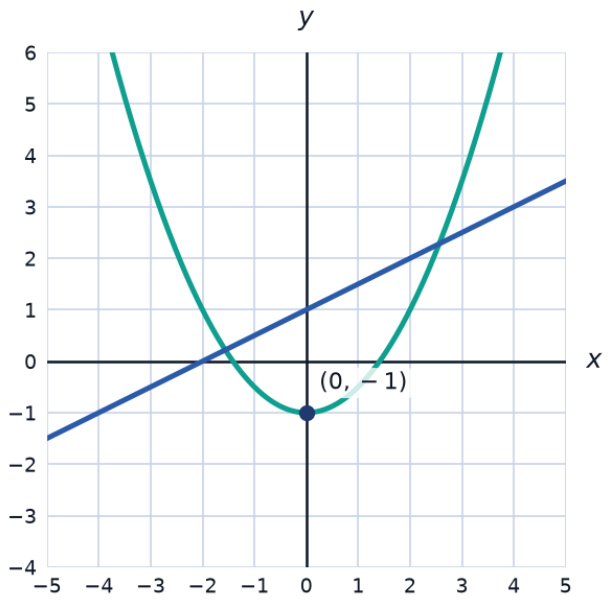
b f_y

c f_{xx}

d f_{xy}

2 [Foundational] • Target: First and second partial derivatives

For $f(x, y) = e^{xy}\sin(x + y)$, find f_x using the product rule.



3 [Foundational] • Target: The gradient and directional derivatives

Find the gradient ∇f of $f(x, y) = x^2y + 3y^3$ at the point $(1, 2)$.

4 [Foundational] • Target: The gradient and directional derivatives

Find the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4 \rangle$.

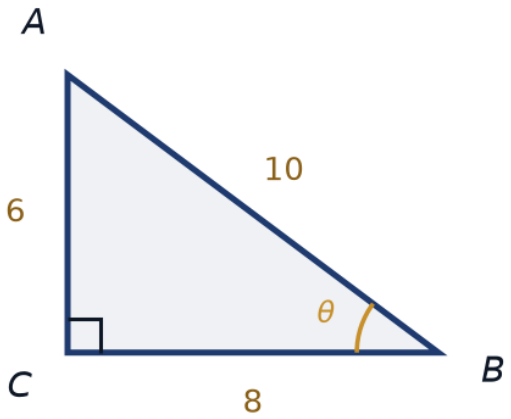
5 [Foundational] • Target: Tangent planes

Find the equation of the tangent plane to $z = f(x, y) = x^2 + y^2$ at the point $(1, 2, 5)$, using $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$.

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6 [Foundational] • Target: Tangent planes

Find the equation of the tangent plane to $z = xy$ at the point $(2, 3, 6)$.



7 [Foundational] • Target: Critical points of multivariable functions

Find the critical point(s) of $f(x, y) = x^2 + y^2 - 4x + 2y + 5$ by setting $f_x = 0$ and $f_y = 0$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Critical points of multivariable functions

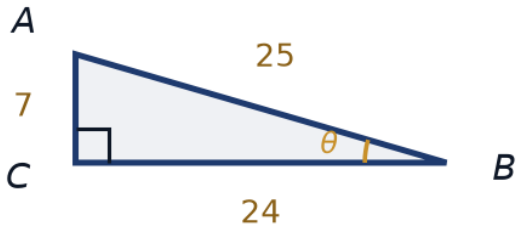
Verify that the critical point found above is a minimum, using the second derivative test with $D = f_{xx}f_{yy} - (f_{xy})^2$.

9 [Intermediate] • Target: The chain rule for multivariable functions

Let $z = x^2y$, where $x = t^2$ and $y = t^3$. Use the multivariable chain rule $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$ to find $\frac{dz}{dt}$ at $t = 1$.

10 [Intermediate] • Target: The chain rule for multivariable functions

Let $w = x^2 + y^2 + z^2$, where $x = \cos t$, $y = \sin t$, $z = t$. Find $\frac{dw}{dt}$ using the chain rule.



11 [Intermediate] • Target: Maximum rate of increase

Find the direction in which $f(x, y) = x^2 + xy$ increases most rapidly at the point $(1, 2)$, and state the maximum rate of increase.

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12 [Intermediate] • Target: Maximum rate of increase

Find the direction of steepest descent for $f(x, y) = x^2 + y^2$ at the point $(3, 4)$.

13 [Intermediate] • Target: Implicit partial differentiation

For the surface $x^2 + y^2 + z^2 = 9$, find $\frac{\partial z}{\partial x}$ by implicit differentiation (treating y as constant).

14 [Intermediate] • Target: Classifying critical points with saddle points

Find and classify the critical point(s) of $f(x, y) = x^2 - y^2$, using the second derivative test.

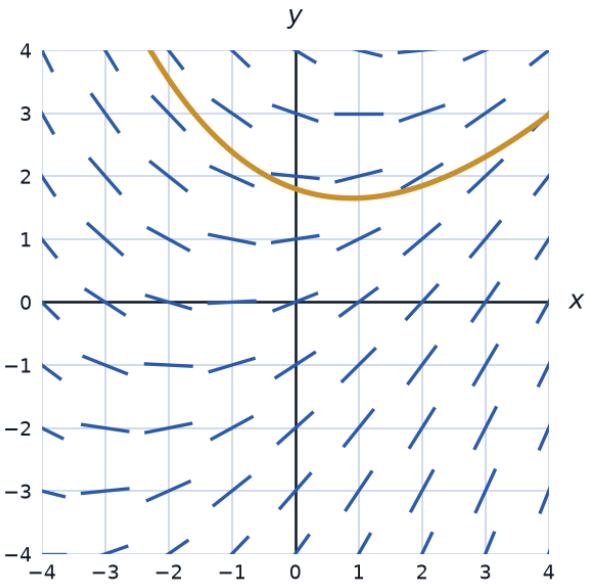
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Classifying critical points with saddle points • Cross-Lesson: 8.02 Volumes of Revolution

Find and classify the critical point(s) of $f(x, y) = x^3 - 3xy + y^3$.

16 [Challenging] • Target: An applied optimization problem • Cross-Lesson: 8.03 Arc Length and Improper Integrals

A company's profit is modeled by $P(x, y) = 100x + 120y - x^2 - 2y^2 - xy$, where x and y are units of two products. Find the production levels that maximize profit by setting $P_x = 0$ and $P_y = 0$.



17 [Challenging] • Target: Higher-order mixed partial derivatives • Cross-Lesson: 9.01 Separable Differential Equations

For $f(x, y) = x^3y^2 + \sin(xy)$, find f_{yx} and f_{xy}, and verify they are equal (as guaranteed by Clairaut's theorem for continuous mixed partials).

18 [Challenging] • Target: Linear approximation in two variables • Cross-Lesson: 1.01 Precalculus Review for Calculus

Use the linear approximation $f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$ to estimate $f(2.1, 2.9)$ for $f(x, y) = x^2y$, using the point $(2, 3)$.

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19 [Challenging] • Target: Linear approximation in two variables • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

The volume of a cylinder is $V(r, h) = \pi r^2 h$. A cylindrical can is manufactured with radius $r = 3$ cm and height $h = 10$ cm, but due to manufacturing tolerances, the radius might be off by as much as 0.05 cm and the height by as much as 0.1 cm. Use the total differential $dV = V_r\, dr + V_h\, dh$ to estimate the maximum possible error in the computed volume.

20 [Challenging] • Target: Linear approximation in two variables • Cross-Lesson: 11.01 Sequences and Series

For $f(x, y) = x^2y - 3xy^2 + y^3$, find each of the following at the point $(1, -1)$:

a $f_x(1, -1)$

b $f_y(1, -1)$

c The gradient vector $\nabla f(1, -1)$

d The directional derivative in the direction of $\langle 1, 0 \rangle$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: First-Order Partial Derivatives

Find f_x and f_y for $f(x, y) = 3x^2y - 5xy^3 + 4x - 2y$.

a. $f_x = 6xy - 5y^3 + 4$, $f_y = 3x^2 - 15xy^2 - 2$

b. $f_x = 6xy - 5y^3$, $f_y = 3x^2 - 15xy^2$

c. $f_x = 6x - 5y^3 + 4$, $f_y = 3x^2 - 15y^2 - 2$

d. $f_x = 3x^2 - 15xy^2$, $f_y = 6xy - 5y^3$

22 [Foundational] • Target: First-Order Partial Derivatives

Find $\frac{\partial z}{\partial x}$ for $z = e^{x^2y}$.

a. $2xye^{x^2y}$

b. $x^2e^{x^2y}$

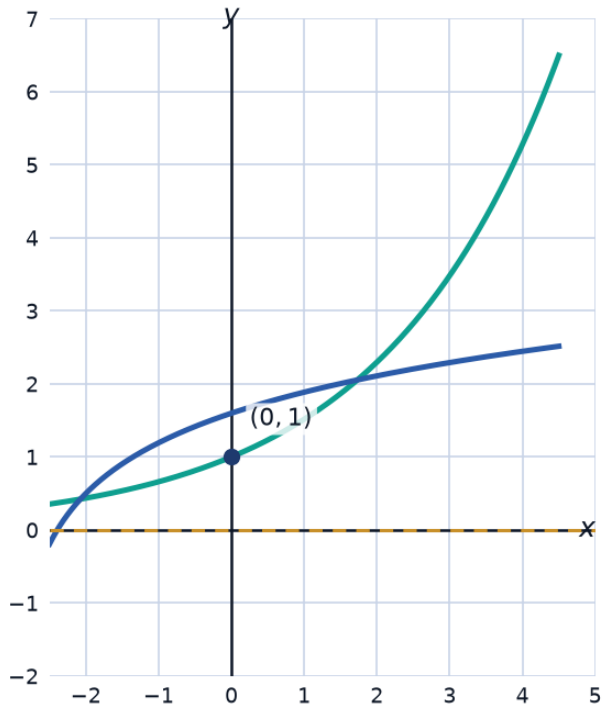
c. $2xe^{x^2y}$

d. $2xye^{2xy}$

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23 [Foundational] • Target: First-Order Partial Derivatives

Find $f_y(1, 0)$ for $f(x, y) = \ln(x^2 + y^2 + 1)$.



a. 0

b. 1

c. 2

d. 1/2

24 [Foundational] • Target: Second-Order Partials and Clairaut's Theorem

What does Clairaut's Theorem (equality of mixed partials) state for continuous second partial derivatives?

a. $f_{xy} = f_{yx}$

b. $f_{xx} = f_{yy}$

c. $f_{xy} = -f_{yx}$

d. $f_{xx} + f_{yy} = 0$

25 [Foundational] • Target: Second-Order Partials and Clairaut's Theorem

Find f_{xx} for $f(x, y) = x^4y^2 - 2x^2y$.

a. $12x^2y^2 - 4y$

b. $4x^3y^2 - 4xy$

c. $2x^4 - 0$

d. $12x^2 - 4$

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26 [Foundational] • Target: Second-Order Partials and Clairaut's Theorem

Verify whether $u(x, y) = e^x \sin y$ satisfies Laplace's equation $u_{xx} + u_{yy} = 0$.

a. Yes, $u_{xx} + u_{yy} = e^x \sin y - e^x \sin y = 0$

b. No, $u_{xx} + u_{yy} = 2e^x \sin y$

c. No, it equals $e^x \cos y$

d. Only when $x = 0$

27 [Foundational] • Target: The Gradient Vector and Directional Derivatives

Find the gradient $\nabla f(x, y)$ for $f(x, y) = x^2 y + 3y^2$ at the point $(2, 1)$.

a. $\langle 4, 10 \rangle$

b. $\langle 4, 7 \rangle$

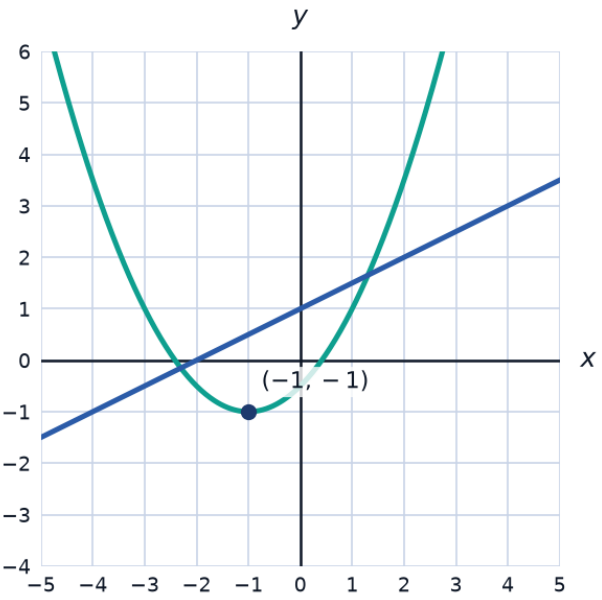
c. $\langle 2, 6 \rangle$

d. $\langle 10, 4 \rangle$

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28 [Foundational] • Target: The Gradient Vector and Directional Derivatives

In which direction does a differentiable function f increase most rapidly at a point P?



a. In the direction of the gradient vector $\nabla f(P)$

b. Opposite to $\nabla f(P)$

c. Orthogonal to $\nabla f(P)$

d. Along the level curve

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Gradient Vector and Directional Derivatives

Find the directional derivative of $f(x, y) = 2x^2 - 3y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4 \rangle$.

a. $-12/5$

b. $12/5$

c. -24

d. 0

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30 [Intermediate] • Target: Tangent Planes and Linear Approximations

Find the equation of the tangent plane to $z = x^2 + y^2$ at the point $(1, 2, 5)$.

a. $z - 5 = 2(x - 1) + 4(y - 2)$

b. $z - 5 = (x - 1) + 2(y - 2)$

c. $2x + 4y - z = 0$

d. $z = 2x + 4y + 5$

31 [Intermediate] • Target: Tangent Planes and Linear Approximations

Use the linear approximation $L(x, y)$ of $f(x, y) = \sqrt{x^2 + y^2}$ at $(3, 4)$ to estimate $f(3.02, 3.97)$.

a. 4.988

b. 5.012

c. 5.000

d. 4.950

32 [Intermediate] • Target: Tangent Planes and Linear Approximations

Find the normal vector to the level surface $x^2 + y^2 + z^2 = 9$ at $(1, 2, 2)$.

a. $\langle 2, 4, 4 \rangle$

b. $\langle 1, 2, 2 \rangle$

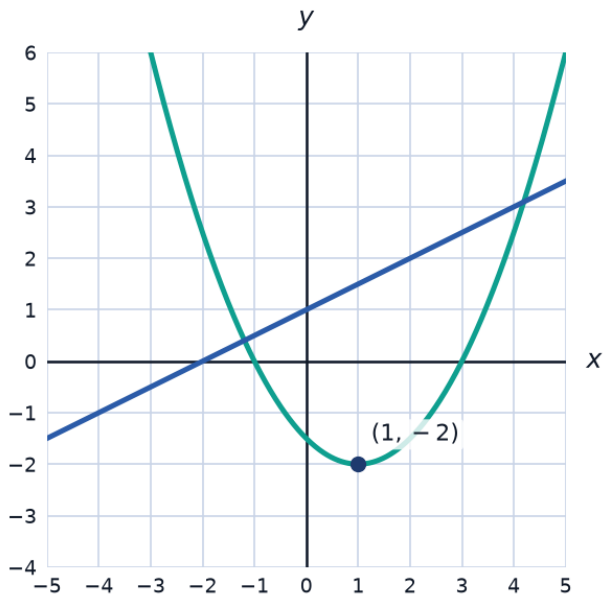
c. Both $\langle 1, 2, 2 \rangle$ and $\langle 2, 4, 4 \rangle$ are valid normal vectors

d. $\langle 2, 2, 1 \rangle$

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33 [Intermediate] • Target: Second-Order Partials and Clairaut's Theorem

Implicit function theorem for optimization: constrained critical points satisfy $abla_{\mathbf{x}} L = 0$ where $L = f - \lambda g$. Name of λ:



a. Lagrange multiplier

b. eigenvalue

c. gradient

d. curvature

34 [Intermediate] • Target: The Gradient Vector and Directional Derivatives

Second-order Taylor approximation:

$f(x,y) \approx f(a,b) + f_x(x-a) + f_y(y-b) + \frac{1}{2}[f_{xx}(x-a)^2 + 2f_{xy}(x-a)(y-b) + f_{yy}(y-b)^2]$. This involves:

a. only first partials

b. only second partials

c. third partials

d. both first and second partials

35 [Intermediate] • Target: Tangent Planes and Linear Approximations

Chain rule with two paths: $z = f(x,y)$, $x = r\cos\theta$, $y = r\sin\theta$. $\partial z/\partial r =$

a. $f_x \cos\theta + f_y \sin\theta$

b. f_r

c. $f_x r + f_y r$

d. $f_x \sin\theta - f_y \cos\theta$

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36 [Intermediate] • Target: First and second partial derivatives

Classify the critical points of $f(x, y) = x^2 - 4x + y^3 - 12y$:

a. $(2, 2)$: saddle

b. $(2, 2)$ min; $(2, -2)$ saddle

c. $(2, 2)$: local min; $(2, -2)$: local max

d. $(2, 2)$ saddle; $(2, -2)$ saddle

37 [Intermediate] • Target: The gradient and directional derivatives

Chain rule: $z = f(x, y)$, $x = x(t)$, $y = y(t)$. Then $dz/dt =$

a. $f_x \frac{dx}{dt} + f_y \frac{dy}{dt}$

b. $f_y dy/dt$

c. $f_x dx/dt$

d. $f_{xx} + f_{yy}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Tangent planes • Cross-Lesson: 1.01 Precalculus Review for Calculus

Find all critical points of $f(x, y) = x^3 + y^3 - 3x - 3y$.

a. $(\pm 1, \pm 1)$

b. $(0, 0)$ only

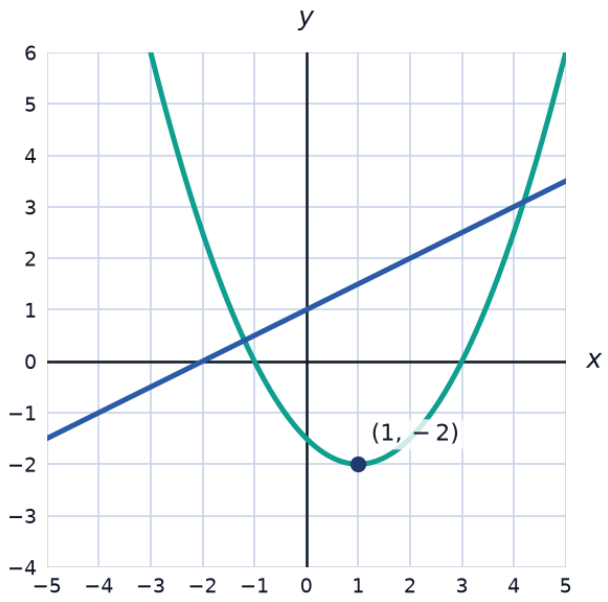
c. $(1, 1)$ only

d. $(1, -1)$ and $(-1, 1)$ only

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39 [Challenging] • Target: Critical points of multivariable functions • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Implicit function theorem: at (a, b), $\partial y/\partial x =$



a. D_xF

b. $-(D_yF)^{-1}D_xF$

c. 0

d. I

40 [Challenging] • Target: The chain rule for multivariable functions • Cross-Lesson: 11.01 Sequences and Series

Second mixed partial: for smooth f, $f_{xy} =$

a. f_{yx}

b. f_{yy}

c. f_{xx}

d. 0

41 [Challenging] • Target: Maximum rate of increase • Cross-Lesson: 12.01 Convergence Tests

Euler's theorem for homogeneous functions: if $f(tx, ty) = t^n f(x, y)$, then $xf_x + yf_y =$

a. f

b. nf

c. $n(n-1)f$

d. 0

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42 [Challenging] • Target: Implicit partial differentiation • Cross-Lesson: 12.02 Power Series and Taylor Series

What is the maximum rate of change of $f(x, y) = x^3 - 3xy^2 - y$ at $(2,\ 1)$?

a. $5\sqrt{10}$

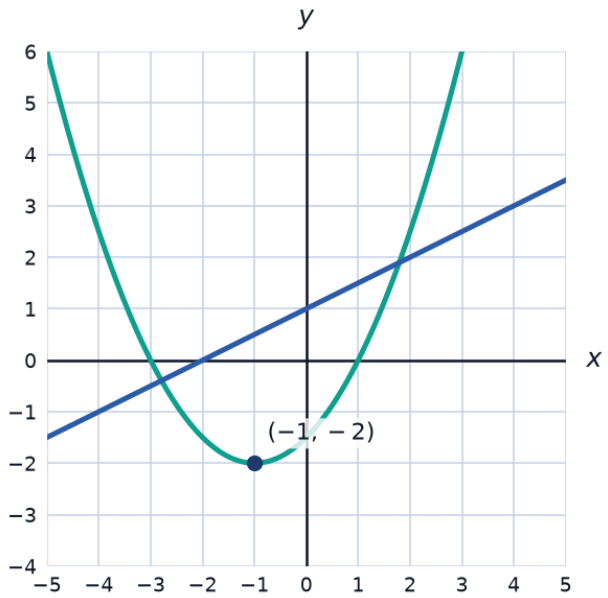
b. 250

c. 22

d. 13

43 [Challenging] • Target: Classifying critical points with saddle points • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

What is the maximum rate of change of $f(x, y) = x^2 - 2xy - 3y^2$ at $(3,\ -2)$?



a. 136

b. $2\sqrt{34}$

c. 16

d. 10

44 [Challenging] • Target: An applied optimization problem • Cross-Lesson: 15.01 Double Integrals

What is the maximum rate of change of $f(x, y) = 3x^3 + xy^2$ at $(2,\ 1)$?

a. 1385

b. $\sqrt{1385}$

c. 41

d. 37

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45 [Challenging] • Target: Higher-order mixed partial derivatives • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Local extrema of f can occur only at critical points (interior) or on the:

a. axes

b. level curves

c. boundary

d. Hessian eigenvalues

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